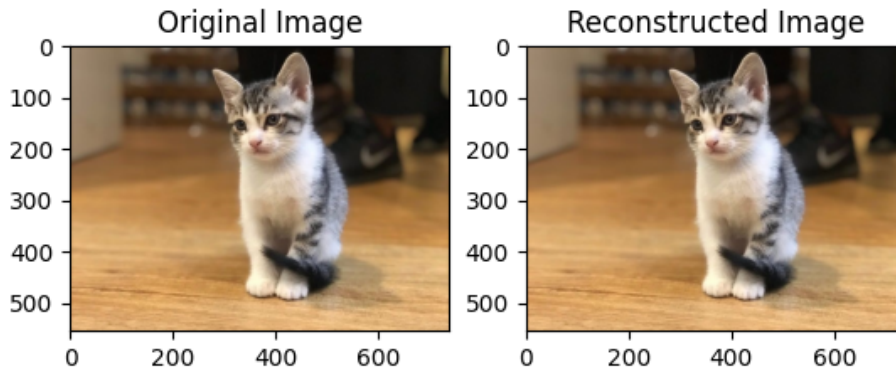


- (1) Write a Matlab or Python code for the 4:2:0 image compression technique.

$B = C420(A)$, where A is the input color image and B is the reconstructed image. Just use the interpolation method for reconstruction. The code should be handed out by NTUCool. (Note: The command `rgb2ycbcr` cannot be used.) (25 scores)



- (2) Why the Mel-frequency cepstrum is more suitable for dealing with the acoustic signal than the original cepstrum? (10 scores)

- ① always real (不必處理複數)
- ② $\sum |x[k]|^2 B_m[k]$ has much less probability to be 0
- ③ The cutoff frequencies of windows match the characteristic of hearing
- ④ DCT (discrete cosine transform) is applied instead of the IFT (使計算量較小)

- (3) Suppose that the cepstrum of $x[n]$ is

$$\hat{x}[2] = 1 \quad \hat{x}[n] = 0 \quad \text{otherwise}$$

Please determine $x[n]$. (10 scores)

步驟：Z transform $\rightarrow$ exponential $\rightarrow$ inverse Z transforms

$$\hat{x}[n] = \begin{cases} 1, & n=2 \\ 0, & \text{otherwise} \end{cases} \xrightarrow{\text{Z transform}} \chi(z) = z^{-2} \xrightarrow{\text{exponential}} e^{z^{-2}} \quad \text{展開泰勒級數}$$

$$\left(\sum_{k=0}^{\infty} \frac{1}{k!} (z^{-2})^k = \sum_{k=0}^{\infty} \frac{1}{k!} z^{-2k} \right)$$

$$\xrightarrow{\text{inverse Z transforms}} \chi[n] = \begin{cases} \frac{1}{n!}, & n=0, 2, 4, \dots \\ 0, & \text{otherwise} \end{cases} \quad *$$

- (4) Suppose that, for a stringed instrument, the frequency of Do is 250Hz. (a) Determine the string length corresponding to Do if the speed of sound at 15°C is considered. (b) What is the string length corresponding to La? (10 scores)

$$(a) f = \frac{340}{2L} = 250 \quad (b) f = \frac{340}{2L} = 250 \times 2^{\frac{9}{12}}$$

$$L = \frac{170}{250} = \frac{17}{25} \text{ m} \quad L \approx 0.4043 \text{ m}$$

- (5) (a) Why a music signal is easier to compress than other vocal signals? (Write at least 3 reasons) (b) Why a cartoon / mark image is easier to compress than other images? (Write at least 2 reasons) (10 scores)

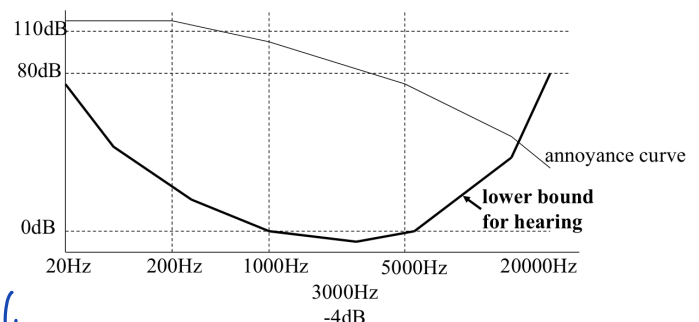
- (a) ① more concentration at the frequencies $f_0, 2f_0, 3f_0, \dots$
 ② for each note, the frequency is fixed
 ③ fundamental frequency are $f_0 \times 2^{\frac{k}{12}}$
 ④ beat, intervals are $T \cdot 2^k$, $k = -1, 0, 1, 2, \dots$ (-拍為T秒)
 ⑤ repeated melody

- (b) ① the color/intensity is fixed within a region
 ② edge can be approximated by lines or arcs

- (6) Suppose that there are three vocal signals (i) $-\cos(1200\pi t)$; (ii) $\sin(5400\pi t)$; (iii) $\cos(20000\pi t)$. (a) Which one sounds the loudest? (b) Which one is the most suitable to sound? (10 scores)

(a) (ii) $\sin(5400\pi t)$

(b) (i) $-\cos(1200\pi t)$



從此圖判斷
 { 人對3000Hz左右最敏感
 { Hz越大, 人耳越感到麻煩

(10 scores)

② DCT 相比於 KLT, DCT independent of the input

② The memory requirement is reduced

(8) Suppose that $P(x = \text{'a'}) = 0.45$, $P(x = \text{'b'}) = 0.3$, $P(x = \text{'c'}) = 0.16$, $P(x = \text{'d'}) = 0.06$, $P(x = \text{'e'}) = 0.03$.

(b) Determine the coding tree of x when using the Huffman code in the binary (二進位) system.

(c) What is the average coding length for each input when using the Huffman code to encode x ? (15 scores)

(b)

0.55 0.45
a
0.3 0.25
b
0.16 0.09
c
0.06 0.03
d e
0.03 0.06

$$= 0.45 \times 1 + 0.3 \times 2 + 0.16 \times 3 + 0.06 \times 4 + 0.03 \times 4$$
$$= 1.89$$

(ended with $(3, 8), (4, 9), (0, 5), (1, 6)$)

Q: Mel-frequency cepstrum 是選取前面幾個 coefficient 做為語音辨識的特徵?

A: 前面 13 個 ($C_x[1], C_x[2], \dots, C_x[13]$)